

Classification of Elements: Periodic Table

1 Introduction to the Periodic Table

The **Periodic Table** is a systematic arrangement of all the known chemical elements based on their **atomic numbers** and the periodic recurrence of their physical and chemical properties. It provides a convenient way to study and compare the properties of elements and helps in predicting the properties of newly discovered elements.

The modern periodic table is based on the **Modern Periodic Law**, which states that:

“The physical and chemical properties of elements are periodic functions of their atomic numbers.”

In the modern periodic table:

- Elements are arranged in the increasing order of their atomic numbers.
- Elements having similar electronic configurations and chemical properties are placed in the same vertical column called a **group**.
- Horizontal rows are called **periods**.
- The position of an element in the periodic table is determined by its electronic configuration.

2 Periodic Properties

The physical and chemical properties of elements show a regular variation with increasing atomic number. Such properties, which vary periodically across periods and down groups in the periodic table, are called **periodic properties**.

The important periodic properties are:

- Atomic Radius
- Ionic Radius
- Ionization Enthalpy
- Electron Gain Enthalpy
- Electronegativity

2.1 1. Atomic Radius

The distance between the centre of the nucleus and the outermost shell of an isolated atom is defined as Atomic Radius.

General Trends

- Atomic radius decreases from left to right across a period due to increase in effective nuclear charge.
- Atomic radius increases from top to bottom in a group due to the addition of new electron shells.

2.2 2. Ionic Radius

The radius of an ion is called its **ionic radius**.

General Trends

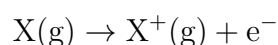
- A cation is smaller than its parent atom because loss of electrons decreases electron-electron repulsion.
- An anion is larger than its parent atom because gain of electrons increases electron-electron repulsion.

Examples

- Na^+ is smaller than Na.
- Cl^- is larger than Cl.
- Mg^{2+} is smaller than Mg.
- O^{2-} is larger than O.

2.3 3. Ionization Enthalpy

Ionization enthalpy is the minimum amount of energy required to remove the most loosely bound electron from an isolated gaseous atom.



Examples

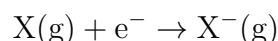
- Sodium has a lower ionization enthalpy than magnesium because its outermost electron is removed more easily.
- Helium has the highest ionization enthalpy due to its completely filled shell.

General Trends

- Ionization enthalpy generally increases from left to right across a period.
- Ionization enthalpy generally decreases from top to bottom in a group.

2.4 4. Electron Gain Enthalpy

Electron gain enthalpy is the enthalpy change that occurs when an isolated gaseous atom accepts an electron to form a gaseous negative ion.



Usually, energy is released during this process; therefore, electron gain enthalpy is generally negative.

Examples

- Chlorine has a highly negative electron gain enthalpy because it readily gains an electron.

- Noble gases have positive or nearly zero electron gain enthalpy due to their stable electronic configurations.

General Trends

- Electron gain enthalpy generally becomes more negative from left to right across a period.
- Electron gain enthalpy generally becomes less negative down a group.

2.5 5. Electronegativity

Electronegativity is the tendency of an atom in a molecule to attract the shared pair of electrons towards itself.

Unlike ionization enthalpy and electron gain enthalpy, electronegativity is not measured directly. It is a relative property.

Examples

- In H–F, fluorine attracts the shared electron pair more strongly than hydrogen.
- Oxygen is more electronegative than nitrogen.

General Trends

- Electronegativity increases from left to right across a period.
- Electronegativity decreases from top to bottom in a group.
- Fluorine is the most electronegative element.
- Caesium and francium are among the least electronegative elements.

2.6 Summary of Periodic Trends

Property	Across a Period	Down a Group
Atomic Radius	Decreases	Increases
Ionic Radius	Generally decreases	Increases
Ionization Enthalpy	Increases	Decreases
Electron Gain Enthalpy	Becomes more negative	Becomes less negative
Electronegativity	Increases	Decreases

3 Classification of Elements into *s*, *p*, *d* and *f* Blocks

Based on the subshell into which the last (differentiating) electron enters, the elements are classified into four blocks, namely the ***s*-block**, ***p*-block**, ***d*-block** and ***f*-block**. This classification is based on the electronic configuration of the elements and helps in understanding their physical and chemical properties. The four blocks together constitute the modern periodic table.

3.1 *s*-Block Elements

- The elements in which the differentiating electron enters the *s*-subshell of the outermost shell are called ***s*-block elements**.
- General electronic configuration :

$$ns^1 \text{ or } ns^2$$

where *n* is the principal quantum number.

- Occupy Groups 1 and 2
- Highly electropositive in nature.
- Group 1 elements are called alkali metals
- Group 2 elements are called alkaline earth metals
- They are placed on the left side of the periodic table

Examples

H, He, Na, Mg, K

3.2 *p*-Block Elements

- The elements in which the differentiating electron enters the *p*-subshell of the valence shell are called ***p*-block elements**.
- General electronic configuration

$$ns^2 np^{1-6}$$

- Occupy Groups 13 to 18.
- Includes metals, metalloids and non-metals.
- Mostly form covalent compounds.
- Contain many elements essential for life such as carbon, nitrogen, oxygen and phosphorus.
- Show great diversity in physical and chemical properties.

Examples

B, C, N, O, S, Cl, Ne, Ar

3.3 *d*-Block Elements (Transition Elements)

- The elements in which the last electron enters the penultimate *d*-subshell are called ***d*-block elements** or **transition elements**.
- General electronic configuration

$$(n-1)d^{1-10}ns^{1-2}$$

- Occupy Groups 3 to 12.
- Located between the *s*-block and the *p*-block.
- All are metals.
- Exhibit variable oxidation states.
- Form coloured ions and compounds.
- Readily form coordination compounds
- Good conductors of heat and electricity.

Examples

Sc, Ti, V, Cr

3.4 *f*-Block Elements (Inner Transition Elements)

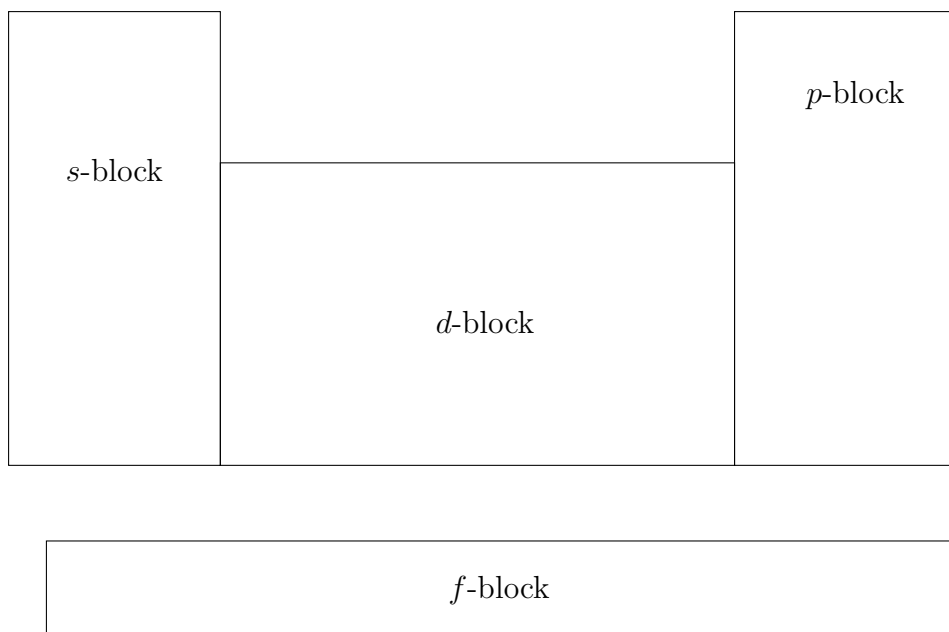
- The elements in which the last electron enters the $(n-2)f$ -subshell are called ***f*-block elements** or **inner transition elements**.
- General electronic configuration

$$(n-2)f^{1-14}(n-1)d^{0-1}ns^2$$

- Placed separately at the bottom of the periodic table.
- Consist of two series:
 - Lanthanoids (58–71)
 - Actinoids (90–103)
- All are metallic in nature.
- Show variable oxidation states.
- Actinoids are radioactive.
- Form complex compounds and exhibit magnetic properties.

Examples

Ce, Pr, Nd, U, Th



4 Periodic Trends in the Properties of Elements

The physical and chemical properties of elements do not change randomly in the periodic table. They exhibit a regular pattern of variation with increasing atomic number. This regular variation is known as the **periodic trend**. These trends arise due to changes in the electronic configuration and effective nuclear charge of the elements.

The important periodic properties that exhibit periodic trends are:

- Atomic radius
- Ionic radius
- Ionization enthalpy
- Electron gain enthalpy
- Electronegativity
- Metallic and non-metallic character

The variation of these properties can be studied across a period and down a group.

4.1 Periodic Trends Across a Period

Across a period, elements are arranged in the increasing order of atomic number. As we move from left to right, electrons are added to the same principal energy shell, while the nuclear charge increases. As a result, the effective nuclear charge increases and influences the periodic properties.

4.2 Periodic Trends Down a Group

Down a group, each successive element possesses an additional electron shell. The increase in atomic size and shielding effect outweighs the increase in nuclear charge, resulting in characteristic trends.

4.3 Reasons for Periodic Trends

The periodic trends in the properties of elements are mainly governed by the following factors:

1. **Increase in Effective Nuclear Charge (Z_{eff}):**

Across a period, the number of protons in the nucleus increases while the added electrons enter the same shell. Consequently, the effective nuclear charge increases, pulling the valence electrons closer to the nucleus.

2. **Increase in Number of Electron Shells:**

Down a group, each successive element has one additional electron shell. The outermost electrons are farther from the nucleus, resulting in an increase in atomic size.

3. **Shielding Effect:**

The inner electrons shield the outermost electrons from the attractive force of the nucleus. The shielding effect increases down a group, reducing the effective nuclear attraction experienced by the valence electrons.

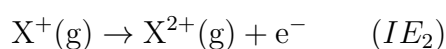
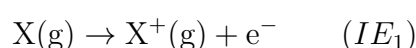
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Electron Gain Enthalpy	Becomes more negative	Becomes less negative
Electronegativity	Increases	Decreases
Metallic Character	Decreases	Increases
Non-metallic Character	Increases	Decreases

5 Ionization Enthalpies IE_1 and IE_2

The energy required to remove the first electron from an isolated gaseous atom is called the **first ionization enthalpy** (IE_1). Similarly, the energy required to remove the electron from a unipositive ion is called the **second ionization enthalpy** (IE_2)

The ionization processes are represented as:



The second ionization enthalpy is always greater than the first ionization enthalpy because after the removal of the first electron, the atom is converted into a positively

charged ion. The number of protons in the nucleus remains the same, but the number of electrons decreases. Hence, the effective nuclear attraction on the remaining electrons increases.

Hence,

$$IE_2 > IE_1$$

Factors Influencing Ionization Enthalpy:

The ionization enthalpy of an element depends on the following factors.

1. Atomic Size

As the atomic size increases, the distance between the nucleus and the valence electron increases. Consequently, the force of attraction between them decreases, making the removal of the electron easier.

$$\text{I.E} \propto \frac{1}{\text{Atomic Size}}$$

2. Nuclear Charge

An increase in nuclear charge increases the attractive force exerted by the nucleus on the valence electrons.

$$\text{I.E} \propto \text{Nuclear Charge}$$

3. Shielding (Screening) Effect

The inner-shell electrons shield the valence electrons from the attractive force of the nucleus. A greater shielding effect reduces the effective nuclear attraction.

$$\text{I.E} \propto \frac{1}{\text{Shielding effect}}$$

4. Electronic Configuration

Atoms having completely filled or half-filled subshells possess extra stability. Therefore, more energy is required to remove an electron from such atoms.

Examples:

- Nitrogen ($2p^3$) has a higher ionization enthalpy than oxygen ($2p^4$).

5. Penetrating Effect

In a given shell, the penetrating power of valence of the valence electrons decreases in the order of $s > p > d > f$. So, IE value decreases in the same order

6 Previous Year Questions

1. What is periodic property? How the following properties vary in a group and a period? 1. Atomic Radius 2. Electron Gain Enthalpy 3. Electro negativity (EN) 4. Ionisation Enthalpy
2. Define IE_1 and IE_2 . Why is $IE_2 > IE_1$ for a given atom? Discuss the factors that effect IE of an element.
3. Write an essay on s,p,d and f block elements.